Unit 36 - Computer Game Platforms and Technologies
Assignment 4 - Game Platforms

Introduction

Scenario / Vocational Context - *You are an Indie Game Designer, as part of your design and development process you would like to explore and understand the different game platforms and devices that your games could be made for.*

As such, your initial research will explore the development of different platforms and their evolution over time.

You should complete a comparison of old and new platform types in the following areas:

- *Gaming PCs (Optionally VR)*
- *Consoles*
- *Handheld Gaming Devices (Optionally AR)*
- *Arcade*

Each comparison should consider performance, capabilities, sound, graphics, games and features. You should consider features and limitations in detail using accurate terminology and writing with clarity. The Focus should be on user experience when playing games.

Each comparison should clearly state what platform type is being considered, use of images is recommended. It is expected that your work be completed in either MS Word or MS PowerPoint.

Task 1

(U36P1/M1/D1) - Describe and comprehensively explain game platform types with some (generally) appropriate, consistent and correct use of subject terminology with reference to detailed illustrative and elucidated examples.

Task description - *To achieve P1:U36 you must describe platform types and consider some limitations. Using some subject and technical terminology.*

To achieve M1:U36 you must explain platform types and their limitations for all your examples. Using illustrative and detailed examples with good use of subject and technical terminology.

To achieve D1:U36 you must comprehensively explain game platforms with accurate and detailed examples of their features and limitations. Using supporting evidence and clear examples. You must also demonstrate consistent and accurate use of subject and technical terminology.

Introduction

On this assignment I will be describing and explaining about each platform type, including: gaming PCs, consoles, handheld devices and arcade machines. Describing the different aspects that has evolved with the different platform types, which may include examples of: performance, capabilities, sound, graphics, games and features found in each platform type.

I will be mentioning the modern state of the platform type and comparing it to the platforms of older generations, seeing how much the platform has developed over time. Giving examples of older devices and newer devices to give evidence for the features and limitations.

Gaming PCs (Optionally VR)

The PC has had a mixed history with video games, which is mostly due to the competition it faced during the generations and the argument if the Personal Computer is a valid video game platform, especially during the era when it seemed that consoles would make PCs obsolete for video games.

In modern day, the PC platform varies in capabilities and performance depending on the different hardware components and manufacturer on said hardware found within the PC. Which is why PCs are usually measured on three different categories: low end, medium end and high end. The reason for this is that not all PCs are fully built by a company, sometimes the user can build their own personal computer by purchasing the required hardware and software.

Medium to high end PCs usually perform much faster than all other gaming platforms, including consoles. So if a user desired performance for a gaming platform, with sufficient research on the topic, PC would be their first choice without a doubt.

An example of an older generation PC performance would be a 90's custom built computer with an Intel Pentium 3 (450 MHz), an ATI Rage 128 Magnum (32 MB VRAM) and 128 Mb of RAM. The hardware on this PC would be primal compared to today's PC standards. However, for the era it was considered very high end since it can run the blockbuster games of the time well above the recommended amount.

An example of a newer generation PC's performance would be an Intel i9-7900X Processor (4.30 Ghz), Nvidia GTX 1080 Ti (11 Gbps GDDR5X VRAM) and 16GB of DDR4 RAM. A gaming PC with this hardware, by modern standards, are typically on very high end of performance, due to the fact it can run any graphically demanding game currently released and even games yet to be released for a significant amount of time.

Essentially, the PC platform has a significant amount of capabilities when compared to other platforms, with the closest one being mobile devices, more specifically smartphones and tablets.

The reason that PCs have such vast amount of capabilities is due to the operating system installed in said personal computer. Whether it is Microsoft's WIndows, Apple's Mac OS or Linux.

Even though other gaming platforms have operating systems on them, they don't come as close to the versatility of the operating system found on personal computers. With Linux OS being the most customisable out of the rest, because it is freely open for improvement and customisation to anyone.

However, there was an era where the operating system on a computer was so limited to a point where computers were initially seen as obsolete for home, work or research use by professionals. That was until computers started developing operating systems, such as MS DOS, eventually coming up with computers that would become a commercial success.

An example of an older generation PC's capabilities is the Apple 2, released in 1977 by Apple. A very early computer that become a commercial success due to its innovative capabilities when compared to other computers of the time.



With the initial ROM operating system then upgraded to the even better Apple DOS 3.3, supported a hierarchical file system with much larger storage capabilities. It gave the computer the ability to be used at office work, research and even for home use. Due to Apple's innovative design of their computers, it helped PCs evolve over time to what is modern day capabilities.

An example of a newer generation PC's capabilities is the Apple iMac, specifically the 2017 version. Even at modern times Apple is a widely successful company, one of the few technology behemoths, making competition against Apple virtually impossible.With this status as a company, it is clear that they have been improving their computers over the generations.



Their iMac computers may not be the first choice for gaming, like custom built gaming computers or prebuilt gaming computers by other companies, but it does have a wide amount of capabilities. Which is why it is still such a commercial success, especially in office work and research, just like the very first computers made by Apple.

With the still improving Mac OS, with incredible security and outstanding user friendliness, it is clear that Apple has realised how to make a well refined computer. Due to the fact they started

off at a good position and fixed the mistakes that appeared in the past, proving that knowledge in PC evolution can make a company very successful.

The same can be said about video games and gaming PC's alike. With the improvement of hardware over the years, has made them much more capable in software versatility and faster performance when rendering heavy programs such as video games.

Gaming on computers started at a very early stage, where one game created in 1962 by a Steve Russell known as Spacewar. This was a locally multiplayer game, used by two controllers, which the users had to try and shoot each other down. Spacewar was only playable on a computer known as PDP-1.

With that creation, they realised that gaming would be a useful hobby for computers, which was then seen as obsolete. It was thanks to inventions like this and the constant development by companies, due to the profitable gaming business that started to grow over the years, that on the modern day we have a vast amount of video games on so many platforms, with PC and consoles being the flagship of development.

Where notable games PC's games such as Doom, the Quake series including Quake 3 and Unreal Tournament, being the best selling games on PC as they evolved. However, consoles at the time were still ahead of PC, due to the fact that consoles were much more user friendly.

Especially with installation of video games, where a console would only require a single CD or cartridge, while a PC game may need over 2 or 3 disks to install a single game. Making it very tedious for PC gamers, sometimes making them move on to consoles to play the same game but with much more user friendliness.

That was until game developer company Valve, headed by Gabe Newell, announced an online, video game shop and download service called Steam. With this creation, PC gaming hadn't been the same since the ZX Spectrum or Commodore 64 were overtaken by consoles, until the release of the software by Valve.



The software was created to standardise one platform for shopping for video games online without the need of physical installment, thanks to downloads. It also allowed for a much easier experience in multiplayer over PC to an international level, due to the friends systems that is included with steam. Which provided an internal community with the software, which is still developing to this day.

Thanks to steam and other software alike developed over the years, the PC platform has successful offline and online games which can be obtained without going to the local game

store. With successful offline single player games that does not require an internet connection, after downloaded to the device, removing the need to play by installing it from multiple discs.

Also with online multiplayer games, available to play with different users around the world, potentially creating new friends and a community thanks to the friends and community system that Valve created on their software.

PC also has very good backwards compatibility when compared to other platforms, which means you can buy all games, whether they are old and more recent, for cheaper prices than they are on the market for consoles.



In comparison, the modern PC is significantly more powerful when compared to a PC from the 90s. That is expected however, since hardware and software is constantly being improved over time. Meaning that even the most powerful modern PCs would degrade from high, to medium and eventually to low end due to these improvements.

When compared to other devices, the PC throughout the eras always had much more capabilities and performance when compared to other devices, but sometimes with a higher price tag, even with the ability of upgrading the hardware and software. However, that it itself was a limitation on older PCs for video games.

The reason that it is limitation for PCs, especially for the older PC, is that it used to be a high end machine during the 90’s. However, it got degraded over the years as technology improved, making it obsolete due to the performance not meeting any modern requirements. Meaning that more money has to be spent upgrading it, which could cost more than other platforms overall, or an entirely new PC needs to be brought.

Due to the constant evolution of technology, which does improve the capabilities and performance, therefore increasing the user experience with video games such as graphics and overall faster rendering of the game.

However, it is also a limitation for the modern PC which will affect it in the long term. Not only because of the absurdly high price required for the hardware components, but the fact that it will be outmatched as technology develops. While it does last longer than other PCs with less performing hardware, the user can still spend much cheaper amounts of money on a console or lower end PC to run the same video games.

Thanks to the development of steam, due to the user friendliness problem of installing games, the prices of PC games were so incredibly cheap in the sales that gaming stores have completely ceased the sales of PC games, due to Steam and other similar software.

Overall, this was the definitive reason that PC gaming began to increase in popularity again and has since been able to reach to a competitive level to consoles. The reason for all this is that PC is a great gaming platform.

However, the lack of user friendliness over the years due to the software of computers and the fact that the only way to obtain video games in the first place was tedious for the user. With software capable of installing through internet access, which is widely available in the modern day, PCs can benefit with ease of installation and upgradability of computer components, which makes the PC last longer than other platforms.

With the improved performance and capabilities, it gives the PC the ability to try out new methods of gaming, such as VR. Even though it is still in early development, VR seems to have a huge recognition in the gaming industry.



Making it possible that VR would become very successful for video games in the future. However that is not guaranteed, since it is still very early into the development of VR to determine the success.

Consoles

Consoles are simple, somewhat portable devices in the shape of a box. Containing the hardware required to run video games offered for said console, but to also be enough to produce in mass amounts, also to be profitable and affordable to the public.

These devices are controlled with an ergonomic handheld device called a controller to access the software of the machine, so it's intentional purpose can be carried out by the user, which is to play the desired video game.

Throughout history, consoles were and still are the leading platform of video games in general, making the platform the flagship of the gaming industry and culture. The reason for this is because they were designed and manufactured for (somewhat) affordable gaming at home comfortably.

It is thanks to consoles, such as Nintendo's *Nintendo Entertainment System*, that video games first started getting popular during the 1980's, which has had a successful increase through the

decades. With millions of children invested in this new electronic era, that would later on expand to a much wider audience.

Through their development, consoles have reached a very high milestone over the years, especially with the new Xbox One X and PS4 Pro console, containing hardware capable of running high resolution games with aesthetically pleasing graphics.

Which isn't very surprising, since consoles with the best performance are usually one of the most sold consoles, that is if it doesn't have other significantly terrible design choices or an absurd price cap.

Furthermore, as the hardware and software improved on consoles, so did video games. Aspects such as the graphics, story writing, sound design and overall budget of video games, due to the increasing market. The capabilities that the consoles had has also been improved, sometimes even going so far to create an entire new generation of consoles.

For instance, an older generation console that had good performance capabilities for the era it was made, is the TurboGrafx 16 with it's revolutionary graphics chip giving it immense performance compared to other competitors. So much that it started a new generation, the fourth-generation.



In today's standards this console is very primitive, but that is a very good aspect, because it shows that consoles have developed well over the years. However, this console at the time was seen as a high end console, because it was extremely powerful compared to the rest of the consoles of the day, meaning that games had more potential to be improved.

However, it was thanks for improvements as vast as this console, that generations have been constantly evolving over the years. Not only increasing the performance and capabilities for consoles, but also increasing the operating system software size and capabilities for user friendliness.

Also increasing the creativity of game developers, giving them a lot more to work with on graphics, sound design, gameplay design, story writing and the length of the game. Which in turn increases the quality of games, that may rise the price of the games up.

That may be a limitation, but as prices for games rise up, the better the consoles and the games for said consoles would improve over the years. However, they may become really good to a point where it may create a video game addiction problem for more users.



This may seem far stretched for some people, but it's becoming a reality with more recent consoles and video games.

For instance, the new Xbox One X, which is the more modern console which is considered to have the best performance on the market. It essentially is an upgraded version of the orginal Xbox One, released late 2013 to early 2014 depending on what country the user is in.

The Xbox One X (PS4 Pro also comes under this category) is a good example of the price of consoles increasing as they develop, with a price of GBP £449 (USD $499). Which is a very high price considering that consoles such as the original PS4/Xbox One. Even the older PS3 and Xbox 360 reaching prices close to this cap while they are released, but not as much the Xbox One X/PS4 Pro, even with inflation in mind.

Another example of prices increasing, are newly released games being released at GBP £49.99 (USD $59.99), such as the newly released Assassin's Creed: Origins for PS4, Xbox One and PC. Which is a AAA game, but in the past AAA games used to be sold for £39.99 ($53.99), such as the release of Grand Theft Auto 5 or Call of Duty: Black Ops 3.

This is both a limitation and an advantage, depending what type of person is being mentioned. For the customer, it's an initial limitation but a long term advantage, since they have to spend more for video games and platforms. However, they will benefit from the improvements later on with video games and consoles.

For companies it is a benefit, but extremely risky. Since the company does get money profit from consoles and video games, giving them the ability to expand on the generation by improving the platform. However, if the company makes a mistake, it won't be taken lightly by the public.

Especially if the public paid a high price for the console or video game. Increasing the chance of the company getting a really bad reputation, which could cause the company to lose profits and do poorly in the market in the future, which would leave them lagging behind competition. Causing the company to go bankrupt or get canceled due to the unrecoverable descent from evolution.

As evidence shows, due to the high popularity and profits that video games bring, the graphics on consoles have definitely been improved over the years, with modern day graphics being seen as "photo realistic", especially with the introduction of



4K resolution compatible consoles such as the Xbox One X and the PS4 Pro.

Even though graphics on a console device isn't as good as what the PC platform is capable of, due to much heavier hardware customisation, the evolution of graphics on console video games have proven to be the most innovative compared to other gaming platforms.

Along with graphics, sound on consoles have been proven to be very important, as the first consoles didn't even have sound compatibility due to the hardware and software limitations of the era. However, with the introduction of sound with the consoles, they proved to be just as important as graphics are.

For instance, the most successful was the console that introduced the 3rd generation: The *Nintendo Entertainment System.*



The introduction of Nintendo's *Nintendo Entertainment System*, or the NES, is regarded as the console to resemble the gaming platforms we know and use today. It was the console that introduced the 3rd generation of consoles, the 8-bit generation.

The 8-bit generation introduced much better hardware with the consoles, meaning that graphics could be improved along with the sound design, giving developers the opportunity to be more creative with graphical and audible design, increasing aesthetic pleasure to the user.

Even though it is seen as primitive to modern standards, the NES supported much higher resolution graphics and larger spectrum of colours compared to the competitors, which was mind blowing for the time.

It was the improved graphical visuals and uncomparable performance that made the console the highest selling console of all time, selling almost 62 million units worldwide. Despite the video game crash of 1983, which saw the demise of essentially all 2nd gen platforms due to cheap imitations and fakes, making the gaming culture untrustworthy of the companies.

Overall, this proves that, even if the gaming market is currently facing poor public view, such innovation could improve that public image. Thanks to the NES, consoles have built a reputation that is now virtually unbreakable if it carries on with the cycle of improvements. Unless there is a significant price increase or lack of innovation, which is possible.

It is seen throughout the generations there were multiple companies that started producing consoles, such as Atari, Magnavox, Nintendo, NEC, Sega, Sony and Microsoft being the most

popular throughout the generations. While the companies that brought innovative devices to the market such as: Nintendo, Sony and Microsoft, are still strong companies to this day.

While companies that made significant and unrecoverable mistakes, such as Sega, Magnavox and Atari, made them lag behind the competition. Forcing them to cancel the continuation on the video game market. Even cancelling the company as a whole, or in Sega's case, giving up on creating consoles and making video games instead.

In terms of consoles doing so well in the market, as evidence shows, consoles have been designed to carry out the same purpose throughout the generations, which was a device capable to play video games that can be sold in mass amounts. However, they way they were built was significantly changed due to the different hardware and software improvements throughout the years.

The reason that they are called generations is that it shows the leap of hardware and technology used by the leading consoles, which would then be equalled by the competition. Essentially, when new technology is developed, each company produce a console better than the last, forming the next generation. This process still continues on the modern day, only with less competitors.

The difference being that they all work fairly similarly with similar titles and a few exclusive titles, due to the innovative companies profiting from their hit products. While companies that was once big, made unrecoverable mistakes that forced them to cancel production due to the rapid increase of the video game culture.

Handheld Gaming Devices (Optionally AR)

A handheld gaming device is a broad category, meaning that it doesn't necessarily have to be made for gaming. Instead, it has to be a device which is able to be used without being connected to external hardware through physical wires (unless the battery needs to be charged), and it needs the sufficient hardware and software to run video games on.

Which is why handheld gaming devices can range from a Nintendo DS, which was made specifically for mobile gaming. To a smartphone or tablet, which wasn't specifically made for gaming, but is compatible enough to run games to a satisfactory level.

Gaming on handheld devices such as a smartphone or tablet allows the user to play video games almost anywhere, depending whether the video game requires an internet connection. There are also a range of handheld devices dedicated for gaming, such as the Nintendo Gameboy, Nintendo DS and the newest being the Playstation Vita.

Mobile devices are very limited when compared to other gaming platforms. The reason being that it's sacrificing high performing hardware for portability. Which is seen as more important on a mobile device, due to user comfort.

Which is why mobile gaming devices have risen on popularity on modern day. Due to the fact that in the past, there were very limited hardware performance, meaning that sacrificing it for mobility would render the device useless.

That was until the Gameboy by Nintendo was introduced in 1989, which was a commercial success from start until end. A fourth generation console which was seen as impossible at the time, but proved to be worthwhile when it was released, due to the fact it was way ahead of its time.

The gameboy also proved that mobile gaming was something worth to invest in by companies, whether it be sooner or later down the generations. While there were attempts at mobile gaming devices before the Gameboy, they were not as successful.



Furthermore, many children were attracted to the Gameboy because it provided with much better mobility than a console and built with a screen on the device, instead of relying on a TV like a console would.

Mobile gaming is proven with modern hardware and how successful it is thanks to the Gameboy for the initial realisation of the success of mobile portability. However, even though it was a massive success, it was still a very early stage for mobile devices to be compared to other gaming platforms of the time. Which made the device's graphics look very primal compared to the CRT monitors of the time, now considered to be outdated.

Especially with the notoriously primal screen that the gameboy featured. With the lack of technology to produce backlit screens, especially to a mass amount. Meant that the Gameboy had to be played in a direct light source in order for it to be used to a satisfying level.

Also, the hardware on the Gameboy is very limited, due to its portability. What made the Gameboy so successful wasn't the hardware being powerful enough to run console games, but the graphics being primal enough to be run on a limited device such as a Gameboy. Which did sacrifice a lot of capabilities, one being the very basic screen manufacturing.

However, it did still attract a massive market of children and other users. Proving that mobility was more important than the graphical display or performance of the device. Which was released by Nintendo and other companies.

Meaning that if they improved the device, while still having the same mobility that the original device had. It would prove to be even more of a commercial success, that's if the price isn't absurd enough to be unaffordable for the average consumer.

Which is when they introduced the Gameboy Color a decade later, in 1998. Due to the fact they had the sufficient hardware to create a device with overall better hardware performance and capabilities, such as an improved backlit screen.



Making the original Gameboy obsolete, because it had the same portability and targeted towards the same audience, which was children and young adolescents.

Even with these improvements, the Gameboy Color seems primal to today's standard of technology. However, the time it was designed proved to be a revolutionary device. Especially with the improved screen, speakers while still keeping the same, friendly profile that the original Gameboy had.

With these improvements, many consumers were much more satisfied with the product when compared to the previous Gameboy. Which was already considered to be a very good product since release.

Due to this continuation of improvements to the Gameboy, the company finally realised that mobile gaming is a very important market to invest in. Where they evolved the Gameboy into an entirely new device, called the Nintendo DS, which now featured a touchscreen.

With the Nintendo DS, the company yet improved the performance, capabilities and features of the device. Ergonomic enough to be used by anyone of any age, a friendly profile which attracts children and featuring a touch screen capability. Which was, again, ahead for its time in 2005.



The games featured in the DS was mostly 3D, which was a massive improvement when compared to the previous 2D game compatible devices.

Additionally, with the 3D graphics style that Nintendo has and still uses, the graphics on the Nintendo DS was very difficult to show age. Due to the fact that they didn't focus on photo realistic graphics, instead going for a cartoon-like aesthetic. Which didn't only attract the children and young teenagers to the games, but it is also very hard to age over time because it wasn't meant to be compared with reality or other realistic graphics.

The DS has not only impressed it's audience, but critics too. Making Nintendo look like a behemoth of a company which cannot be compared to to a technological and competitive level. Thanks to this great success of mobile devices over the decades, Nintendo decided to make mobile gaming one of it's priorities. Which is still apparent today, with the Nintendo DS platform still being continued and the relatively new Nintendo Switch.

Also with the success of mobile devices being so successful to Nintendo, other companies such as Sony realised that this was an important market to invest in, since Nintendo was yet to have a solid competitor.

Therefore, Sony introduced a new mobile gaming device year after the release of the DS.



The Playstation Portable, or PSP, was a great success because it astonished people with the graphics being better than the Playstation 1, even sometimes being comparable to the Playstation 2. Which attracted both children and a more mature audience because the PSP featured a wider variety of games, some targeting different audiences.

The reason that the PSP was successful was the same reason that Nintendo's mobile gaming device was popular, the mobility of the device giving the user the ability to use it wherever they are. Meaning that there is a mass satisfaction with the mobility of devices.

Which was also released by Apple years later, after the release of touchscreen devices such as the iPod, the iPhone and then the iPad. Even though these devices aren't essentially gaming devices, they have the hardware and software sufficient enough for video games to be played on.



WIth the creation of the App Store and games started being developed on mobile devices which was brought by a lot of users, Apple and the game developers released the increase in downloaded video game apps on these devices.

Realising the factor that stands out on mobile devices, which are the capabilities of gaming on a mobile platform due to the convenience. As most people carry their phones with them everywhere they go, it is easier for those people to play games wherever they are, depending on the type of game.

Unlike consoles or PCs, they don't have to set it up in one place, which was the exact same reasons that Nintendo's and Sony's previous mobile gaming devices were so popular. This fact

has been apparent from the day the Gameboy was released up until the modern day, with smartphones, tablets and more advanced mobile gaming devices.

However, even with this success of mobile devices, a rising limitation of gaming on mobile devices are that some games need online to access certain features or even the whole game, depending how it functions. Especially with the increasing reliability that the public has with wireless connection to the internet.

Also the fact that games found in mobile devices are generally downloaded through some sort of virtual store, where the user can browse for different games. Which does let the user browse the type of game the user wants, but requires a stable internet connection to browse and download the video games.



The cost of the games on mobile phones were very cheap, especially when compared to other gaming platforms. For instance, in modern phones, popular mobile games are mostly free, but have the ability to purchase in-game packages. Meaning that the user only has to access the App Store, download the game with an internet connection and then use the game wherever they are, especially if it is an offline type game.

Even if the user has to pay for a game, they usually pay at most £5. It still is significantly less than video games found on consoles and PCs, yet it still provides many hours of fun for the user. Which is another reason that mobile gaming is desirable on the modern day.

Also, games on the mobile are nowadays usually played by utilising the touchscreen, which makes it very easy to access by people of all age groups. This type of interaction is extremely user friendly, as long as the touchscreen is responsive enough to be satisfying to the user.

However, earlier versions of mobiles which played very simple games, such as the popular snake and brick breaker, would just use up to 5 buttons on the keypad up, down, left, right and, in the case of brick breaker, the middle button to fire. Which shows that the improvements to mobile phones have proven to evolve to a much better standard, not only to mobile consoles such as the DS or the PSP.

Overall, the success of mobile gaming devices wasn't due to the fact that they have had very high performance or extremely advanced hardware or software, but it was the fact they are mobile in the first place. However, it does not mean that a device cannot be of high performance or contain advanced features because it will help push through the competition, especially at an affordable price.

This is why the Gameboy, even though extremely primal with hardware technology, had the software sufficient enough to be run on the limited hardware, making it very successful. Which is

also why that updates on older smartphones and tablets slow them down, due to the software being more sophisticated that the hardware of older devices cannot support it to a recommended level. That in itself is a limitation if the user doesn't want to upgrade the device.

Even with all these advantages, there is still room for improvement. The reason that mobile gaming can also be very limited, is that the games can drain the battery life, meaning that the user has to charge the device, removing the mobility aspect of the device.

Another limitation is that the games are only played with simplistic controls, especially on smartphones and tablets. Even though touch screen are very user friendly, the controls of the game is designed by the game developers, instead of the manufacturer of the device. Meaning that the user will experience different types of video game interaction, whether it is a good control mechanism or an awful mechanism.

Lack of internet connection is another limitation, due to the fact when there is no games that use online features, some user are forced to use mobile data to access the video game. This in turn can then use up all of the users mobile data, such as 3G or 4G, instead of the unlimited data cap of their broadband.

With the improvements of performance of recent mobile devices, it has given speculation whether augmented reality would become a success in the future. However, augmented reality is still within its testing phase, so not much can be said about it. So the only details that could be said about AR are theories and opinions.

Personally, I believe that augmented reality would become a success in the future with the correct technology, but I think that virtual reality would overshadow augmented reality for video games.



The reason I believe this is because virtual reality immerses the player into the gaming world, as a form of escapism. Which I believe is more important than immersing the game world into the real world when dealing with video games.

Arcade

Arcade machines are simple, coin operated machines that have been around ever since the 1960's and most popular in the 80's. Since then arcade machines have appeared in many public places such as restaurants, bars and amusement arcades. Even to the modern day, but not as much as it was years ago and there are reasons for this.

At first, arcade machines seems like a very good concept, which it was during the time where consoles and other platforms were only owned by those who could afford it. However, on the modern day, arcade machines seem like a waste of money when compared to something an individual could afford and own such as a console, mobile device or PC.

Since arcade machines are made to be played for an average of 50p a game, it means that if the players charter dies in game, the player will have to buy a second life. Also, games on arcades are different depending on what the machine is advertising the the public. Meaning that the game on this platform is only restricted to what is advertised.

This was one of the biggest reasons why arcade machines have become extremely unpopular compared to what it was during the 80s and 90s. Another reason is due to the other gaming platforms improving over the years in hardware, software and price. Basically making arcade machines almost obsolete.

Since arcade machines featured very little development compared to the other platforms, most arcade machines feature low performance; Which is very insignificant when compared to other gaming platforms, especially PCs and consoles.



Although, there are modern arcade machines feature a console, such as a PS4 or an Xbox One, instead of the arcade machine containing the hardware itself. However, these type of arcade machines are usually found in game hardware stores, to persuade the user to buy the specific console found inside the arcade machine.

With fully built arcade machines today, they still contain the same hardware and games found on arcade machines 20 to 30 years ago, meaning that almost no development was made to arcade machines.

For example, In the modern day, there are some arcade machines that have 3D graphics. However, there are still arcade machines with the retro-style 2D arcade games found in public areas. Which can attract a certain audience that are intro the retro-style of video games or those who want to find a nostalgic experience.

However, this audience is very limited when compared to the rest of the public. Which explains why arcade machines started to get very unpopular, as other gaming platforms evolved while arcades stayed relatively the same as the day they were introduced.

While During the 1980's, when arcades were popular, vector based arcade games started to appear these used geometric lines to create the images on screen. Also in 1980's laserdisc arcade games become common as they use pre recorded video for the graphics, which made it much cheaper to produce and have better looking graphics than vector based graphics.

Relatively speaking, the graphics on arcade machines are sometimes found to be similar to those found 20-30 years ago. Sometimes there are games that existed at the golden age of arcade games, such as Pac Man, Dig Dug, Street Fighter, Space Invaders, Donkey Kong and Mario.

However, even though the arcade machine didn't see any evolution over the years, the games that appeared with them became widely successful and well built games. Which is probably why arcade machines haven't really evolved.

The evolution of the revolutionary games began when the arcade machine first became a concept, was when the first game was released with an arcade machine, called "Galaxy Game". Which was the first game to be released with a coin-operator as well. It was placed and installed into Stanford University, the game was produced by Bill Pitts and Hugh Tuck.

The game cost roughly $20,000 to build and programme. The following year the game Pong (a tennis game) was released for the public. Then in the late 70s early 80s the game Pac-Man was released into the public and was hugely successful.

At the start of the golden age of arcade machines, when they were still considered revolutionary in 1978, saw one of the most popular games to hit Arcade games known as Space Invaders.



Where the user must destroy the little aliens before they reach earth. Space invaders became very popular as it was satisfying to play and there was a score system, which was saved this gave people the determination to beat other people's scores.

The success of this game may have been the addiction that people may have with competition, as it was difficult to beat some scores due to the fact that players only had a certain amount of lives to do it in.



Pac-Man was also one of the most popular games to play in the 80's as it was fun to play but was very difficult. The reason that Pac Man was also popular is due to the fact that the use felt that they got their money's worth, as there were over 250 levels and for the user to play, when compared to other games where you only have a level and handful of lives with a single payment.

Also, as Space Invaders, Pac Man and other popular, well made arcade games paved a path for the arcade machine, it became clear that arcade games weren't going to die down anytime soon. Populating the

public areas, whether it be a shopping centre or a restaurant. Which is why nobody really saw the need to improve arcade machines.

At the time, machines like this brought in large amounts of people as all arcade were in public places where they would get more attention. Which may be an advantage to those who love to meet new people and make new friends, but a disadvantage to those who are very introverted and get nervous with public interaction.

As evidence shows, arcade machines from all eras have limitations which is that they cannot access online features and can only have up to 4 multiplayer games which then costs 4 times more than when playing single player.

Also, the attract mode, which is a video of a gameplay walkthrough, that these machines have has a limitation as it can cause screen burn, which is where an image is burned into the screen if it has been left on for to long, causing the machine to be out of use.

Even though it was widely successful, the arcade genre of gaming is not as widely used today as it was 20-30 years ago. This is mainly because home consoles such as the Xbox One and the PS4 are a lot cheaper than the consoles, compared to the console price to average income ratio from back when arcades were popular. Essentially, most people stopped using arcades when home consoles became more affordable.

Which is why I believe that the future for arcade doesn't look good. As home consoles become more affordable to a lot of the public. The user would usually decide to stay at home instead of going to public areas just to play video games.

Overall, an arcade machine is a very simple gaming platform, hence the very little evolution process. I believe that arcades are also becoming obsolete, with other platforms, such as the PC, being able to have emulators that can run the popular games usually found on arcades.

However, the lack of evolution of the machine may be it’s very own limitation, the reason being that there wasn’t a lot of room for improvement over the decades since the golden age of arcade games. Which is why I believe that the future for arcade gaming looks bad, since there are better developments in consoles, PC and mobile gaming platforms.

Such as online multiplayer and games of higher quality being released on those platforms. Essentially, I believe that less and less people are going to the arcade as they have better games at home.

On the other hand, I also believe that arcade machines would prove to be a great advertisement technique in video game retailers, a machine with a console they are selling inside the machine

gives the user a good first hand trial of the console, increasing the chances of the user purchasing the console.

Conclusion

In conclusion, all different types of gaming platform types each have their own advantages and limitations, considering how differently they operate and target the audience from each other. To conclude each different platform type, I will do a simple but factual conclusion on each platform for india game design and development.

For PC gaming, the advantages of making a video game on this platform is that: The graphics have a lot of priority due to the high performance of this platform. It is relatively easy to use to those with experience. Indie games are usually cheap on a PC platform, meaning that a wider audience is more likely to buy it.

The disadvantages is that it is expensive not only to get a high end PC, but also to design and develop a game for a high end PC. More prone to encounter many errors due to PC games usually being larger than the other platforms. Some games won't work depending on specifications of computer, since the platform is heavily customisable. Also, an internet connection is needed in order to download games.

For Console gaming, it is initially cheaper than PC gaming due to the cost of the machine, which is good for indie games since they are also cheap. The console market is huge at the moment, which is both an advantage and disadvantage because there's a wider audience but also more competition. Games optimised for the console tend to work more reliably than a PC while also maintaining very good to decent graphics.

The disadvantages to consoles is that the games may be cheap for indie games, but not as cheap as a PC indie game. Also the fact that modern consoles require memberships to play online, which adds on to cost. While the graphics are very good to decent, the graphics aren't really as good as PC.

For the mobile gaming platform, the mobility allows the user to play the game anywhere they desire, as long as there is internet connection or if the game doesn't require one. It's possible to play with other people through multiplayer, just like other platforms with the ability to communicate with other people by ringing or texting.

Also, some of the older arcade games can be played on mobile, because graphics don't have to be a priority. However, the graphics tend to be decent at best, which isn't a bad thing. The mobile device also offers many potential game choices due to improvements over the years.

The disadvantages to mobile gaming is that games tend to use a lot of battery, reducing the playing time if the user isn't charging the phone. Games also tend to be quite slow if not optimised correctly, since the hardware on mobile devices aren't as good as a console or PC's, mostly due to the fact that games can take up a lot of memory and hard drive space. However, even if the game was well optimised, sometimes a good phone is needed to play the graphically satisfying games.

Lastly, arcade games tend to provide a lot of fun for a low cost, but per turn. It also tends to get people to socialise and make new friends. If the player is good at playing video games, a single turn could last a long time.

The disadvantages to arcade games is that due to the hardware limitations, the graphics don't tend to be very good, which might not matter for some people however. It also requires the user to travel to a specific area just to play the video game and that some people just don't like socializing, whether they are introverted or just uncomfortable playing away from home.

Due to the fact that the player would need to travel, there is a possibility that there would be a queue for some of the most popular games. Furthermore, if the player really enjoys the game or spent a good amount of time on it, there aren't any saves, and depending on the game it can finish very quickly.

Overall, I believe that mobile gaming platforms seems to contain the most potential for an indie game developer, due to the high popularity and mobile gaming benefiting over video game graphics. Also, creating a game for a mobile device tends to be easier than the rest due to the visuals not being a priority.

With PC being another potential gaming platform due to the versatility of the PC, giving a developer huge potential on creating a game to express creativity and gain valuable experience. Which is very important for an indie developer to gain experience like this, since it gives them a higher chance of creating a hit game.

While consoles are possible, but they are a more limited version of a PC. Since the developer can express creativity and gain experience, but not to the degree a PC could. Also the fact that games on console tend to be more expensive and with the online membership costs adding on to the expenses of the user.

With arcade at the least likely choice, because they tend to be unpopular and as limited as a mobile device. Especially, without the important mobility of a device to overshadow the limited graphics or performance of the game. Meaning that not as much experience or recognition will be gained to the developer.